

Assignment 3

Ex-Post ML techniques for Portfolio Optimization

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Assignment III - Report

DA6701 - Data Science and AI in Finance

Part I: Covariance Estimation and Ledoit-Wolf Shrinkage | Part II: Hierarchical Risk Parity (HRP)

1 | Objective, Universe, and Experimental Setup

This report evaluates two portfolio construction frameworks on a 12-stock NSE universe: (i) covariance estimation and regularization effects in a maximum-Sharpe setting, and (ii) Hierarchical Risk Parity (HRP) using correlation-distance clustering and recursive risk allocation. The data window spans **January 2020 to December 2024** for Part I (monthly returns) and **January 2020 to December 2024** for Part II (daily returns). For both LW shrinkage and HRP, the train/test split is 1st four years (2020-2023 for train) and last year (2024 for test).

2 | Part I (LW): Covariance Stability and Out-of-Sample Behavior

The Part I notebook compares sample covariance and Ledoit-Wolf (LW) shrinkage in a maximum-Sharpe construction. The numerical diagnostics show clear conditioning gains under shrinkage:

- Returns covariance condition number: **97.87**
- Ledoit-Wolf covariance condition number: **26.33**
- Shrinkage intensity: **0.1661**
- Conditioning improvement factor: **3.7x**

Matrix-level differences remain meaningful but controlled (mean absolute difference = 0.0394, max absolute difference = 0.1895, Frobenius norm reduction 0.83 → 0.71, relative change = 16.7%).

Part I Out-of-Sample KPI	Returns Cov	Ledoit-Wolf Cov
Annual Return	13.00%	16.03%
Sharpe Ratio	2.737	5.333

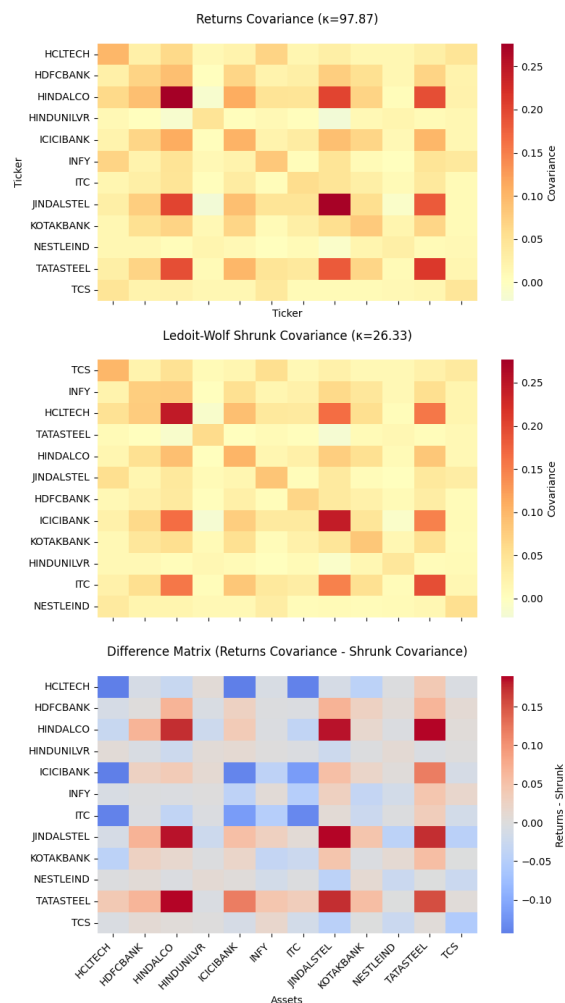


Figure 1: Part I (Viz 1): Sample covariance, LW shrunk covariance, and their difference matrix.

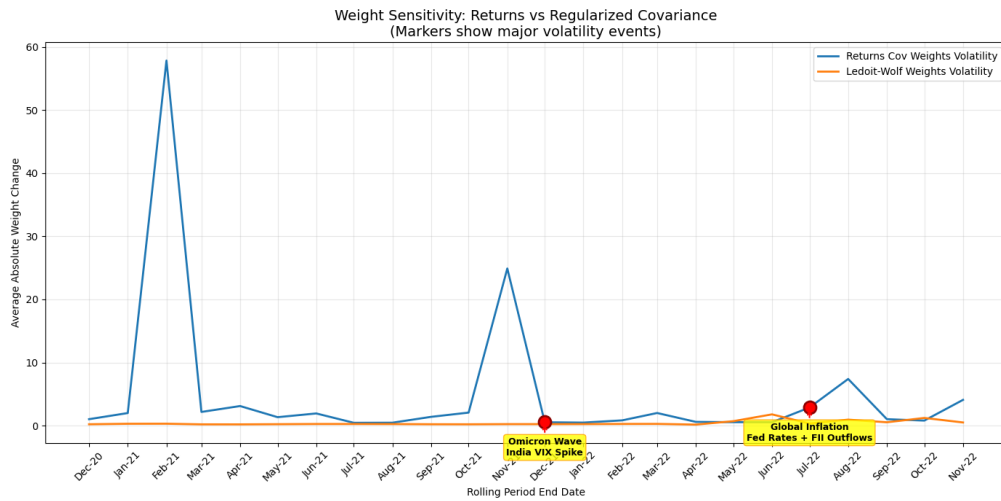


Figure 2: Part I (Viz 3): Weight sensitivity over rolling periods.

Interpretation: Visualization 1 confirms that shrinkage moderates noisy off-diagonal structure while preserving broad covariance blocks. Visualization 3 shows that sample-covariance-derived allocations are highly unstable during stress windows (notable spikes around Feb-2021 and Nov-2021), whereas LW-driven allocation updates remain materially smoother. Together, this supports LW as the more robust estimator for practical, repeated re-allocation workflows.

3 | Part II (HRP): Clustering, Allocation, and Benchmark Comparison

HRP uses the correlation-distance transformation $D_{ij} = \sqrt{0.5(1 - \rho_{ij})}$, single-linkage hierarchical clustering, quasi-diagonalization of covariance, and recursive bisection allocation. At distance cut **0.50**, the five economically interpretable clusters are:

Cluster	Members	Avg Intra- ρ	Volatility Range
1	HINDUNILVR, NESTLEIND	0.608	23.2%–24.7%
2	HCLTECH, INFY, TCS	0.683	25.2%–29.0%
3	HINDALCO, JINDALSTEL, TATASTEEL	0.745	39.3%–50.0%
4	HDFCBANK, ICICIBANK, KOTAKBANK	0.641	28.4%–33.9%
5	ITC	1.000	27.3%

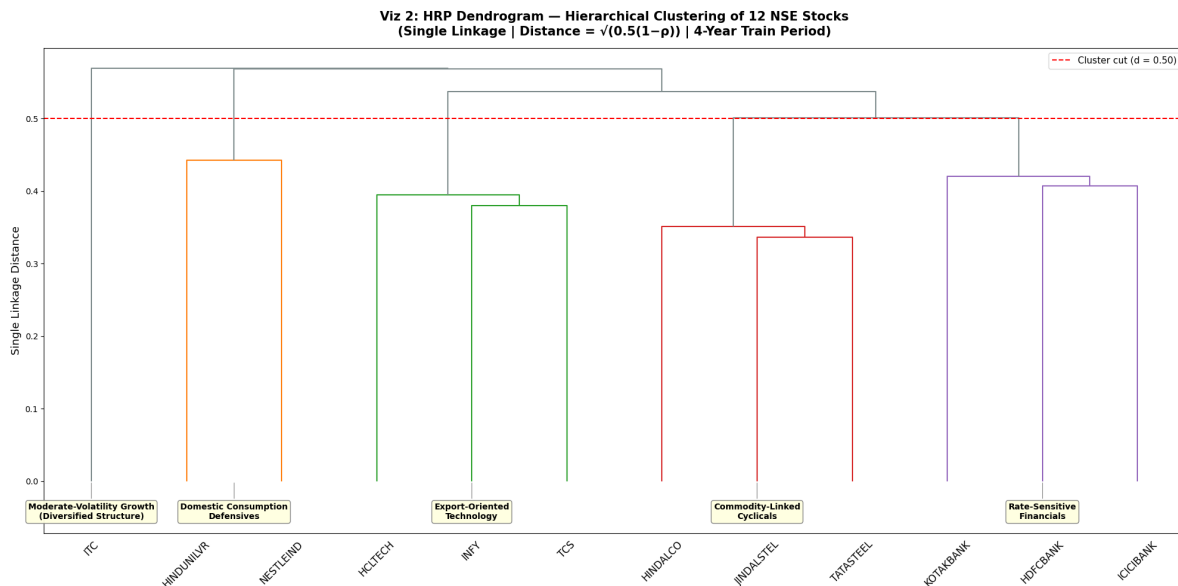


Figure 3: Part II (Viz 2): HRP dendrogram with distance cut $d = 0.50$.

HRP Weights (descending): ITC (16.32%), NESTLEIND (14.12%), HINDUNILVR (12.43%), HCLTECH (11.02%), KOTAKBANK (9.72%), TCS (8.50%), HDFCBANK (6.87%), INFY (6.45%), ICICIBANK (4.80%), HINDALCO (4.60%), TATASTEEL (3.20%), JINDALSTEL (1.98%).

MPT In-sample (train) reference: Annualized return = 32.21%, annualized volatility = 22.73%, Sharpe = 1.1310, with concentrated exposures (notably HCLTECH, ITC, JINDALSTEL, NESTLEIND, TATASTEEL).

2024 Test-Set Metric	HRP	MPT
Cumulative Return	5.97%	17.30%
Annualized Return	6.12%	17.75%
Annualized Volatility	12.13%	15.03%
Sharpe Ratio	0.1379	0.8298
Maximum Drawdown	-10.11%	-8.94%
Avg Monthly Turnover	0.035671	0.039527

4 | Consolidated Insights and Conclusion

Part I takeaway: Ledoit-Wolf shrinkage substantially improves covariance conditioning and produces better out-of-sample risk-adjusted behavior in the notebook experiment, with visibly more stable allocation dynamics under stress.

Part II takeaway: HRP delivers an intuitive diversified risk structure and marginally lower turnover, but in this specific 2024 test sample, concentrated MPT exposures produced stronger absolute and Sharpe performance.

Overall conclusion: covariance regularization is clearly beneficial for estimation stability; HRP offers robust structural diversification and implementation practicality, while realized return leadership remains regime-dependent and can favor concentrated optimizers over short out-of-sample windows.
